

System Identification Report

Topic: Dynamic Modelling of a Sensor Integrated Remotely Operated Underwater Vehicle for Environment Modelling and Mapping of Seabed

Aerial/Underwater Dynamics (ROV System)

1. System Selection and Approval

The selected system is a 6-DOF Aerial/Underwater Dynamic System (ROV) under Category B. This system has been approved by the course instructor for modeling and simulation.

2. Real-World Application

The system represents underwater vehicles such as ROVs and AUVs used in ocean exploration, underwater inspection, and autonomous navigation. The dynamics are governed by thrust, drag, buoyancy, and gravity.

3. System Definition

3.1 Inputs

$u = [\text{surge, sway, heave, roll, pitch, yaw}]$

These inputs are converted into forces and torques using a thruster-based control allocation system.

3.2 Outputs

Position (x, y, z)

Orientation (roll, pitch, yaw)

Linear velocities (u, v, w)

Angular velocities (p, q, r)

Energy (kinetic, potential, total)

Sensor outputs (sonar distances)

3.3 State Variables

$X = [x, y, z, \phi, \theta, \psi, u, v, w, p, q, r]$

The first six represent position and orientation, while the remaining six represent linear and angular velocities.

3.4 Key Parameters

Mass = 73.89 kg

Fluid density $\approx 1000 \text{ kg/m}^3$

Drag coefficient

Buoyancy factor

Forces: buoyancy, drag, angular drag

3. Degrees of Freedom (DOF)

DOF = 6

Translational: Surge, Sway, Heave

Rotational: Roll, Pitch, Yaw

Justification:

The ROV operates in a three-dimensional fluid environment requiring full spatial motion. All six axes of motion are independently controlled using a multi-thruster configuration, enabling complete maneuverability in translation and rotation.

5. Key Assumptions

In-compressible fluid – Water density is constant

Rigid body – No deformation occurs

Quadratic drag – Drag $\propto$ velocity²

Constant mass and inertia – No variation during motion

Fixed center of buoyancy – Acts at a constant point

Ideal thrusters – Instant response

No external disturbances – No currents or waves

Simplified hydrodynamics – Added mass ignored

Controlled environment – Motion within bounded region

FBD HAND DRAWN:

